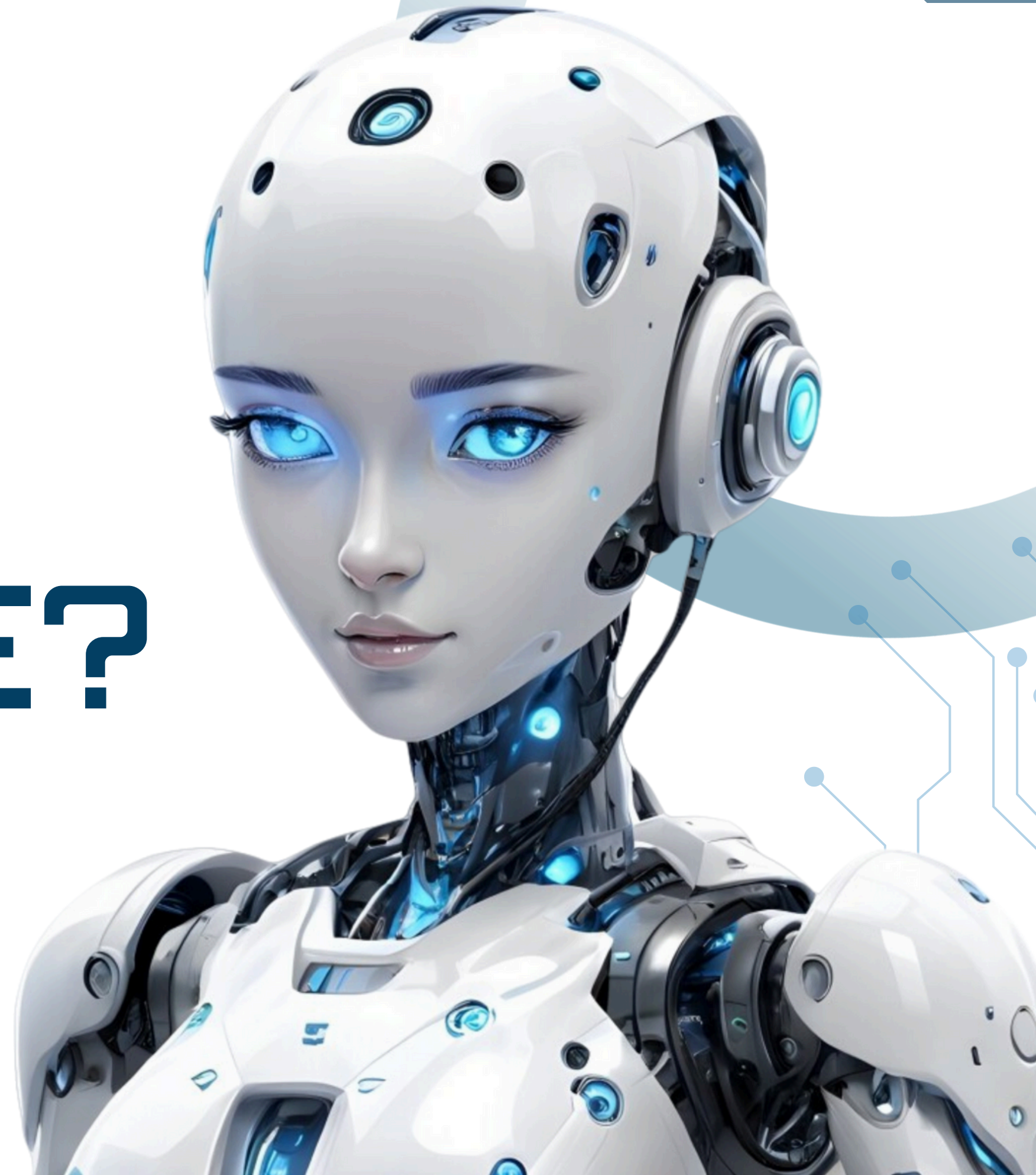


WHAT IS ARTIFICIAL INTELLIGENCE?

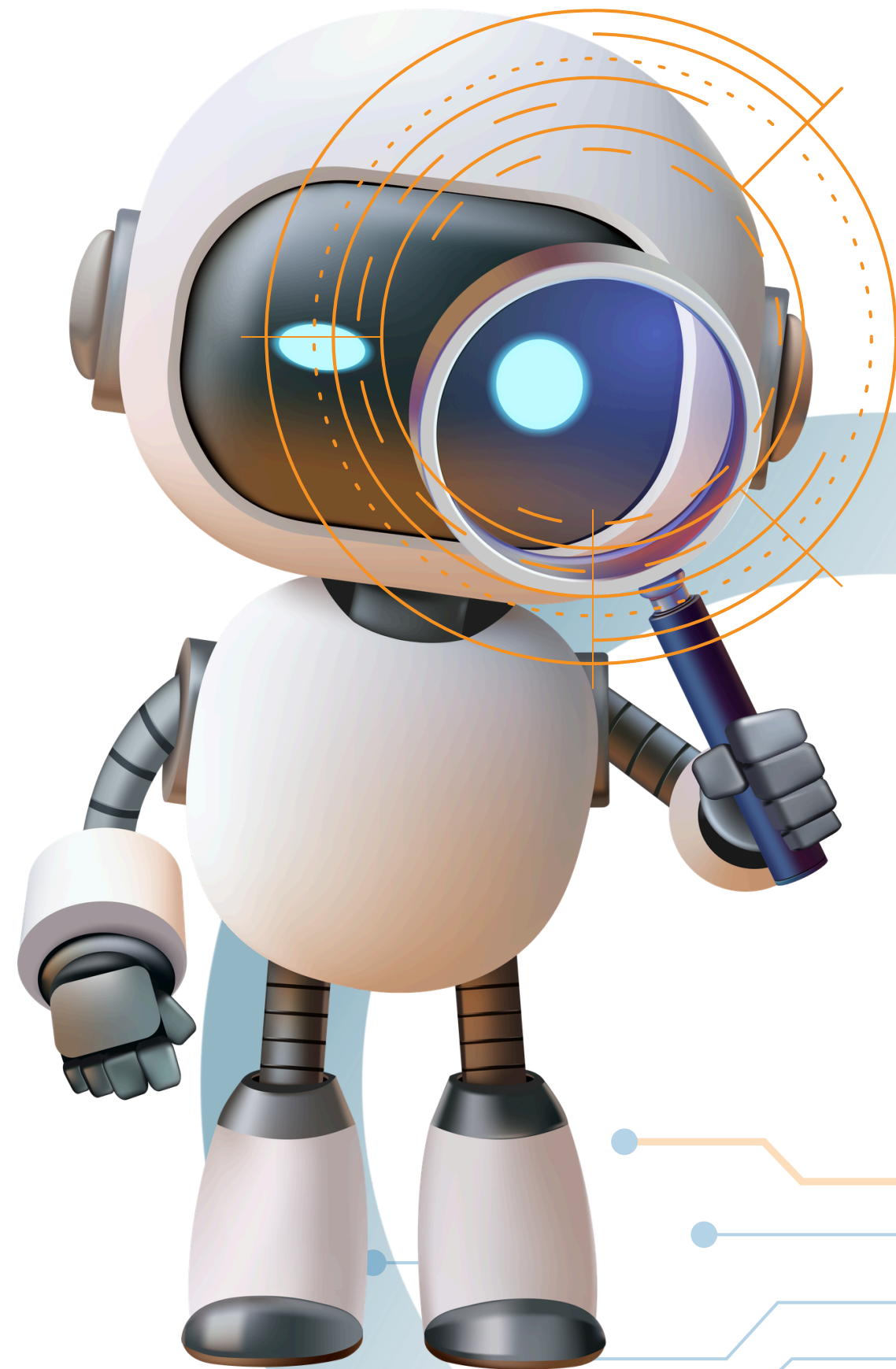


AI ...



Artificial Intelligence refers to computer systems that perform tasks which typically require human intelligence, such as:

- Recognizing patterns
- Making predictions
- Classifying information
- Generating text, images, or recommendations
- Supporting or automating decisions



AI ...



AI systems do not think, feel, or understand in a human sense.

They process data, identify patterns, and produce outputs based on learned statistical relationships.

Key clarification:

AI is not about consciousness — it is about decision influence.



AI VS TRADITIONAL SOFTWARE



Traditional Software

- Follows explicit, human-written rules
- Behavior is predictable
- Logic is transparent
- Same input always produces the same output

Example:

“If the temperature is above 30°C, turn on the air conditioner.”



AI VS TRADITIONAL SOFTWARE



AI Systems

- Learn patterns from data
- Behavior can change over time
- Outputs are probabilistic, not deterministic
- Internal logic may be difficult to explain

Example:

A system that predicts whether a loan applicant is “high risk” based on thousands of past cases.





ETHICAL SIGNIFICANCE

AI systems do not simply execute instructions —
they generalize from past data.



NARROW AI VS GENERAL AI



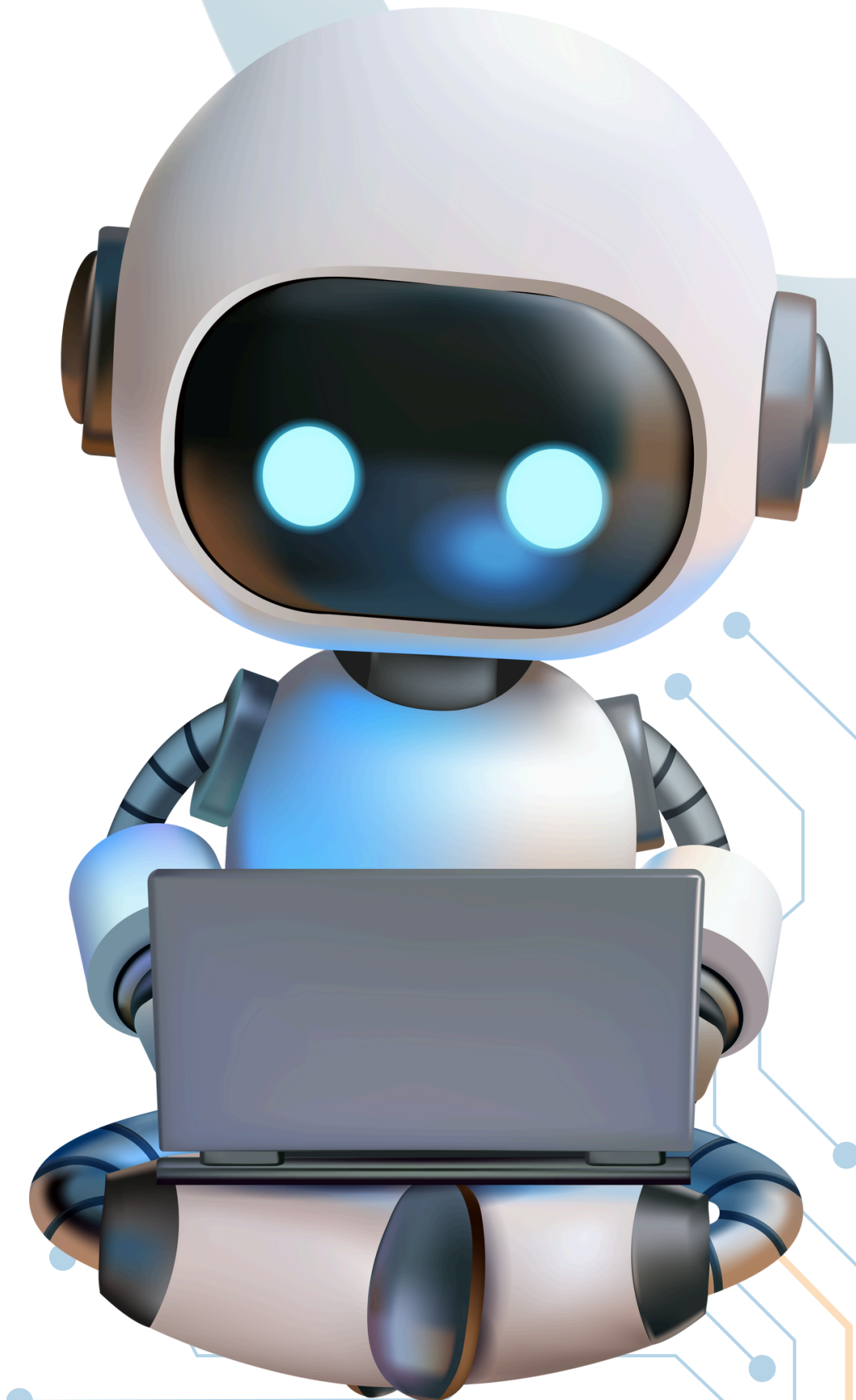
Narrow AI (Today's Reality)

- Designed for specific tasks
- Operates within defined boundaries
- Cannot transfer knowledge between domains

Examples:

- Recommendation systems
- Facial recognition
- Fraud detection
- Language models

All AI systems in real-world use today fall into this category.



NARROW AI VS GENERAL AI

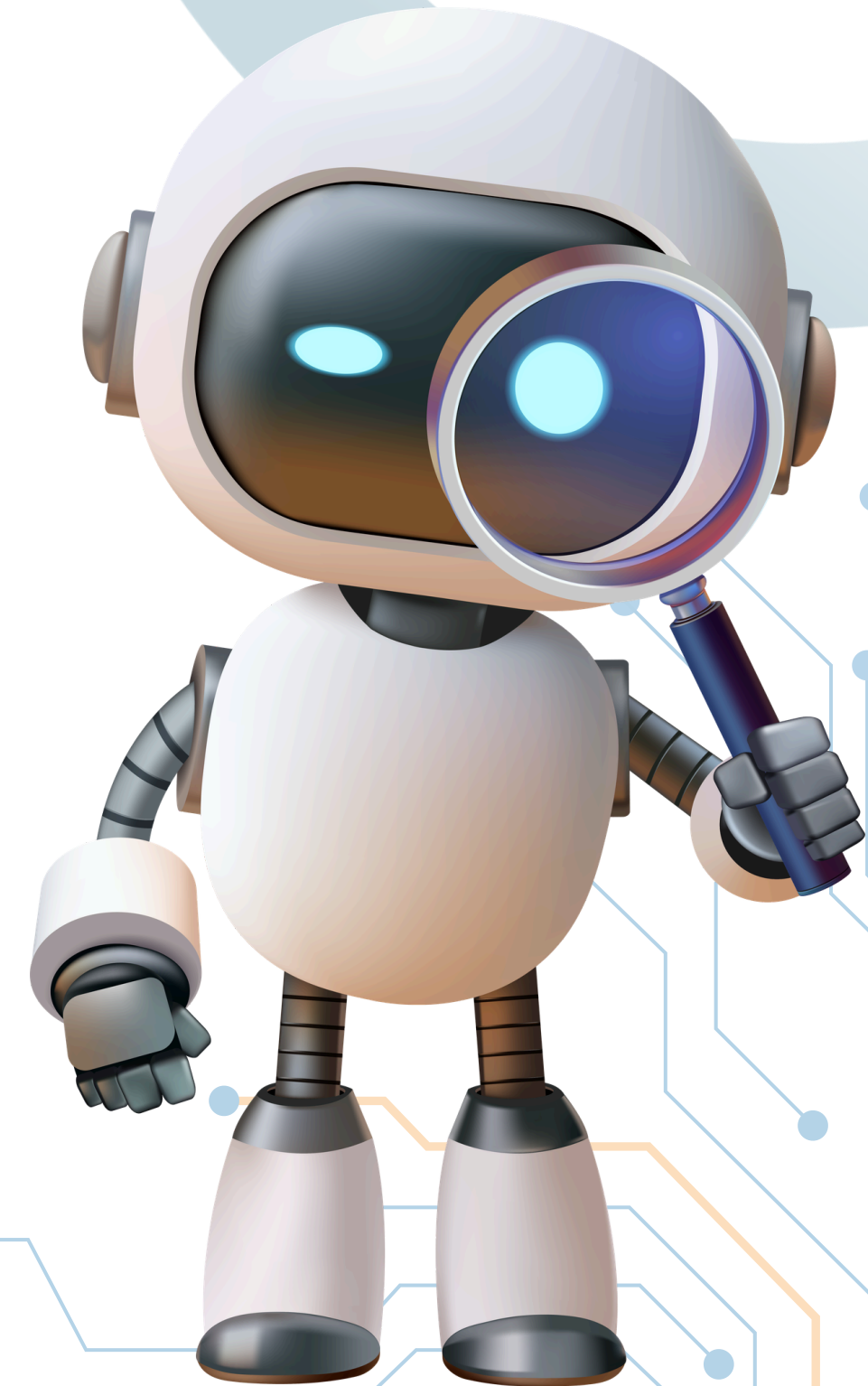


General AI (Hypothetical)

- Would possess broad, human-like intelligence
- Capable of reasoning across domains
- Not currently achievable

Important for this course:

AI ethics today focuses almost entirely on narrow AI systems already deployed.



WHAT AI DOES ...

In real-world settings, AI systems are typically used to:

- Rank people or content
- Score risk or likelihood
- Filter options
- Predict future behavior
- Influence human decisions



WHAT AI DOES ...

AI often does not make final decisions, but it:

- Strongly shapes outcomes
- Influences human judgment
- Operates at massive scale

Example:

An AI system may not “reject” a job applicant, but if it never shows their résumé to a human, the outcome is effectively the same.



WHY THIS MATTERS ETHICALLY

AI systems are ethically significant because they:

- Affect large numbers of people simultaneously
- Can reinforce existing inequalities
- Operate invisibly
- Create power asymmetries between those who deploy AI and those affected by it

Ethical concerns arise not from AI's intelligence, but from:

- Scale
- Speed
- Opacity
- Delegation of judgment



COMMON MISCONCEPTIONS ...



- Misconception 1: “AI is neutral.”
 - Reality: AI reflects the data, assumptions, and goals chosen by humans.
- Misconception 2: “AI is objective.”
 - Reality: AI systems encode value judgments, even when they appear technical.
- Misconception 3: “AI replaces human responsibility.”
 - Reality: AI redistributes responsibility but it does not remove it.
- Misconception 4: “AI decisions are more reliable than humans.”
 - Reality: AI can amplify errors and bias at scale.



ETHICAL FRAMING



Throughout this course, AI will be treated as:

- A socio-technical system
- Embedded in organizations, markets, and power structures
- Designed and governed by humans
- Capable of causing harm without intent

This framing is essential for meaningful ethical analysis.



SUMMARY

- AI systems learn patterns from data rather than following fixed rules
- Most AI today is narrow, task-specific, and already deployed
- AI influences decisions at scale, often invisibly
- Ethical concerns arise from how AI is used, not from intelligence itself

